Learning Charge–Light Correlations: A Multi-Modal Foundation Model for Liquid-Argon TPCs

Paper plan and initial draft (v0)

Haiwang Yu et al.*

Draft v0, November 2026

How to read this draft. Black text is meant to survive into the paper (in rough form). Red marks plan items, TODOs and placeholders. Figures and tables are PLACEHOLDERs from the tech notes, each with a note on how the final version differs. Appendix C is the campaign plan: decisions, the Section-2 job plan (≤200 node-hours), timeline and venue.

Abstract

Foundation models (FMs) pretrained with self-supervision are changing how detector data are analysed, but the FMs proposed for liquid-argon time projection chambers (LArTPCs) see only the ionization charge. LArTPCs also record scintillation light, which gives the event time and an independent calorimetric view. We present a multi-modal FM that reads low-level charge hits from three wire planes and the waveforms of 32 photomultipliers as one token sequence, and is pretrained without labels on MicroBooNE open data. On five tasks – ν hit identification, ν/cosmic event classification, flavour, energy and charge→light prediction – light helps every model, but learning the charge–light correlation *during pretraining* (early fusion) is what matters: the same backbone pretrained on charge alone, with light added afterwards by a strong cross-attention head, recovers only part of the gain. The correlation is learned from 2D wire-plane data, without an explicit 3D reconstruction. We compare four FM recipes – a Transformer with masked autoencoding, a Transformer with a joint-embedding predictive objective, a Point Transformer V3 with self-distillation, and a Mamba-2 state-space model – with light in pretraining at matched size and compute. [**TODO:** one sentence with the Section-2 outcome.]

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*Prepared with the assistance of Claude (Anthropic). Code: `https://github.com/HaiwangYu/fm4npp-lartpc`. Tracking: `docs/paper-campaign.md` and GitHub issue #7. Every figure and number marked PLACEHOLDER comes from tech notes 1–4 (`docs/technote-round{1..4}.pdf`) and will be replaced by the campaign runs.

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1 Introduction

Outline (about 1 page). References from `docs/survey-round1.pdf`.

1. **FMs work for detector data.** Self-supervised pretraining on unlabelled detector data gives reusable representations that reduce label needs and transfer across tasks: masked particle modelling on jets [1, 2], PoLAr-MAE [3] and Panda [4, 5] on LArTPC charge, FM4NPP on a TPC [6], heterogeneous neutrino detectors [7]; review in [8].
2. **Multi-modal FMs are better where modalities are complementary.** Masked multi-modal modelling (MultiMAE [9]) and science omnimodal models (AION-1 [10]) learn cross-modal structure; in HEP, tokenized any-to-any models map between levels of the reconstruction chain [11, 12]. Native early-fusion models do at least as well as late fusion and are better at small sizes [13].
3. **The gap: LArTPC FMs are charge-only.** The scintillation light carries the event time t_0 (which fixes the drift coordinate) and an independent energy measure; conventional reconstruction uses it only after 3D reconstruction, through flash matching [14]. To our knowledge no LArTPC FM pretrains on charge and light together.
4. **Contributions.**
   - The first multi-modal (charge + light) self-supervised FM for LArTPCs, on low-level data (hits after signal processing, optical hits) with no high-level reconstruction.
   - A controlled study showing that the gain comes from correlations learned in pretraining, not from having light at the input (early vs late fusion; shortcut-controlled metrics).
   - A compute-matched comparison of four FM recipes with light in pretraining (Section 4).
   - Analysis of how the correlation is learned: probes, fusion depth, an oracle 3D input (Section 5).
   - A five-task benchmark on MicroBooNE open data with Pandora references. **[TODO:** decide whether to release splits/code as a benchmark.]

2 Data, tasks and models

Data. MicroBooNE BNB inclusive and ν_e CC overlay samples (NoWire HDF5 open data, Zenodo 8370883 and 7261921): simulated neutrinos on real off-beam cosmic data, about 831k events. Inputs: reconstructed wire hits of the three planes (after signal processing and hit finding) and optical hits of the 32 PMTs in a 24 μs window. No Pandora output is used as model input. Splits: 90% pretraining pool, labelled pools of 0.1%, 1%, 10%, 100%; fixed validation and test subsets (TN1).

Tasks and metrics (updated from the tech notes to avoid trivial learning).

Table 1: Downstream tasks. Changes from the tech notes in red.

Task	Target	Headline metric
T1	per-hit ν vs cosmic	per-event F1 of ν hits, unweighted median over events (up-weights small, hard events; TN4 used charge-weighted)
T2	event: in-AV ν / out-of-AV ν / no ν	AUC in-AV vs no-ν on events passing the 200 PE beam-flash cut (TN4), where flash presence alone gives only AUC 0.69
T3	flavour: ν_μ CC / ν_e CC / NC	macro-F1 (as TN1)
T4	log E_ν for CC, fiducial	resolution on a truth-based clean subset, the same for every model: CC, fiducial vertex, ν carries $\geq$[**TODO:** X]% of the event charge, leading lepton contained [**TODO:** finalize]
T5	charge $\rightarrow$ 32 PMT $\times$ 17 light features	test MSE in log(1 + PE); total-PE resolution on $\geq$200 PE events

Models. Each model = tokenizer + backbone + SSL objective + downstream heads, about 20M parameters.

- **qlfm-M CL** (main): charge cells (16 wires $\times$ 64 ticks) and 32 PMT tokens in one sequence; 12-layer Transformer, $d = 384$; masked autoencoding (MAE [15]) over both modalities. **qlfm-M C**: identical, pretrained on charge only.
- **PTv3 + Sonata**: Point Transformer V3 [17] with Sonata self-distillation [18], the recipe of Panda [4], size-matched (21M). In the paper: "PTv3 (as used in Panda)"; we do not use Panda's weights.
- **Pandora** [14]: the reconstruction in the open data, used directly (ν slice, ν score) and through a BDT on 32 Pandora features (+ 8 light features).
- Section 4 adds Transformer + JEPA [16] and Mamba-2 [19] (FM4NPP recipe [6]), each with light.

Heads. Frozen backbone + head; single-task fine-tune (TN1–4); new: multi-task fine-tune, one backbone and five heads trained together, which is how an FM would be deployed.

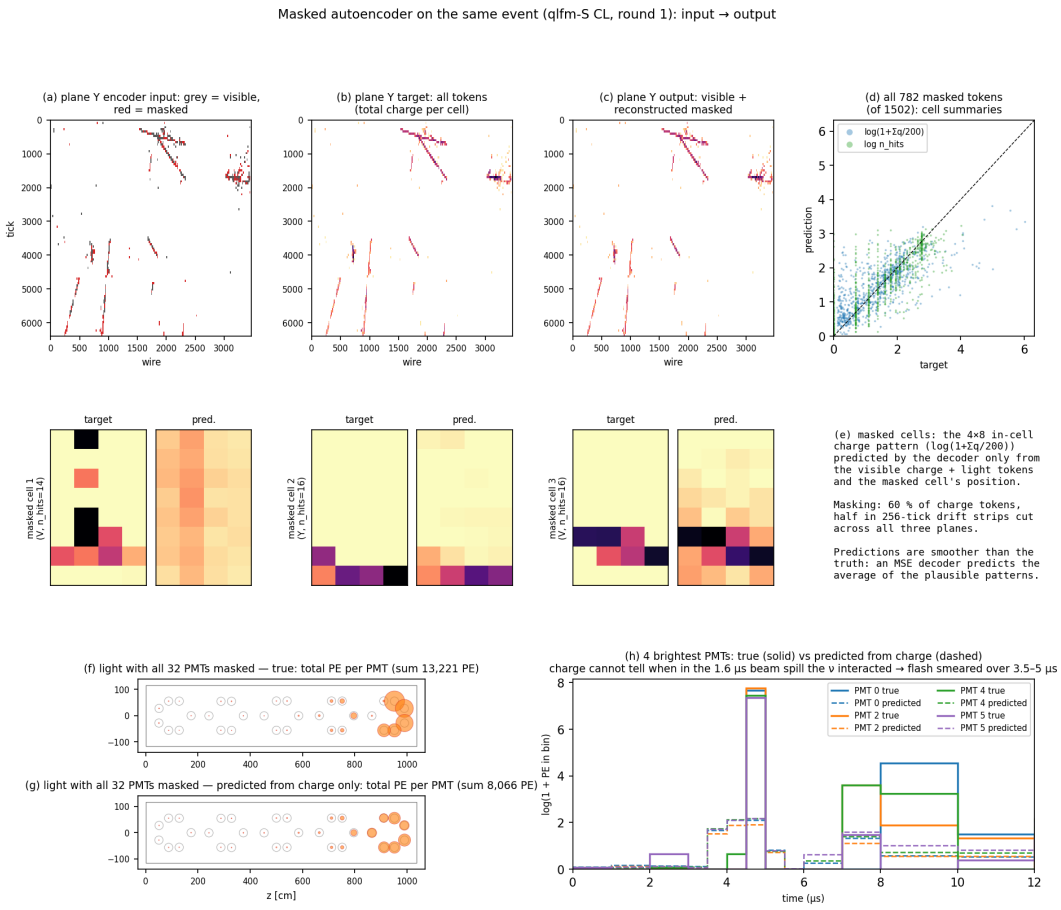


Figure 1: PLACEHOLDER (TN1 Fig. 5). Masked-autoencoder input and output for one event. Final: model overview – charge cells and PMT tokens → shared Transformer → MAE decoder / task heads.

3 Light helps, and learning the correlation is key

3.1 Pretraining works and light helps

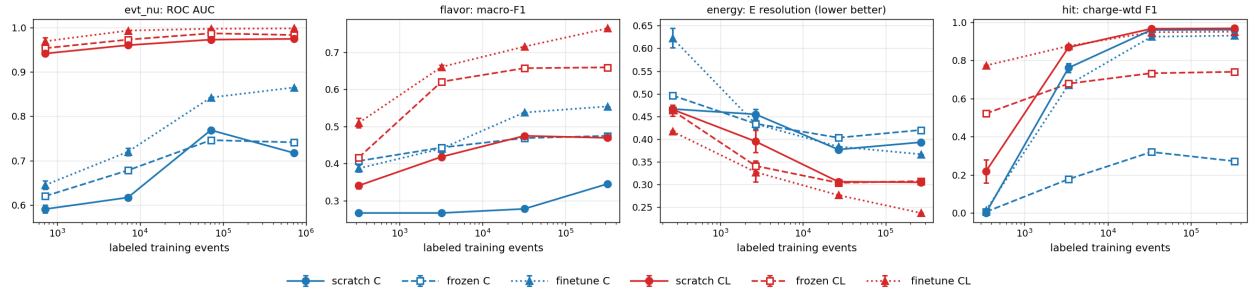


Figure 2: PLACEHOLDER (TN1 Fig. 6, 6.6M models). Test metrics vs labelled events; blue charge only (C), red charge + light (CL); scratch, frozen and fine-tuned. Final: qlfm-M CL vs qlfm-M C only (21M), the updated metrics of Table 1, five panels (T1–T5), frozen and multi-task fine-tune.

3.2 Correlations from pretraining beat light added afterwards

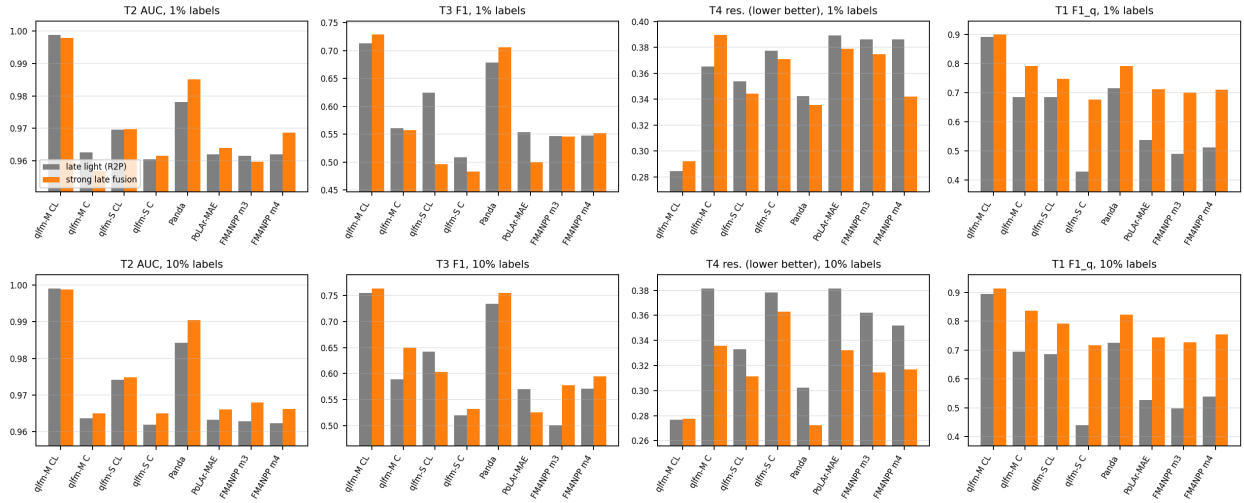


Figure 3: PLACEHOLDER (TN3). Frozen backbones at 1% (top) and 10% labels (bottom) with weak late light (grey) and strong late fusion (orange, cross-attention from charge to PMT tokens). qlfm-M CL has light in pretraining; all other backbones are charge-only. Final: keep, restrict to qlfm-M CL, qlfm-M C, PTv3 + Sonata; updated metrics.

Table 2: PLACEHOLDER values from TN2–4 (frozen, 1% labels, seed 0; T1 charge-weighted median per-event F1; T2 on the 200 PE pass sample; T4 on all CC fiducial events). Late fusion = strong cross-attention head. Final: 0.1% and 1% labels, frozen and multi-task fine-tune, updated metrics, 3 seeds; C-only diagnostic rows (light removed at inference).

Model	Light	T1 F1 ↑	T2 AUC ↑	T3 F1 ↑	T4 res. ↓	T5 MSE ↓
qlfm-M CL	pretraining	**0.91**	**0.99**	**0.713**	**0.284**	**0.503**
qlfm-M C	late head	0.80	0.82–0.86	0.558	0.390	0.586
PTv3 + Sonata	late head	0.79	0.95–0.97	0.706	0.336	0.577
Pandora (+ BDT)	BDT features	0.81	0.95	0.559	0.400	n/a

3.3 Shortcut control and charge→light prediction

The large-light cut (T2) and the charge-only T5 task test whether the model uses charge–light consistency rather than the presence of a flash. qlfm-M CL predicts the PMT light from charge alone, and its own MAE decoder does so with no labels at all (MSE 0.506 vs 0.503 with a trained head; charge-only encoders 0.58–0.59).

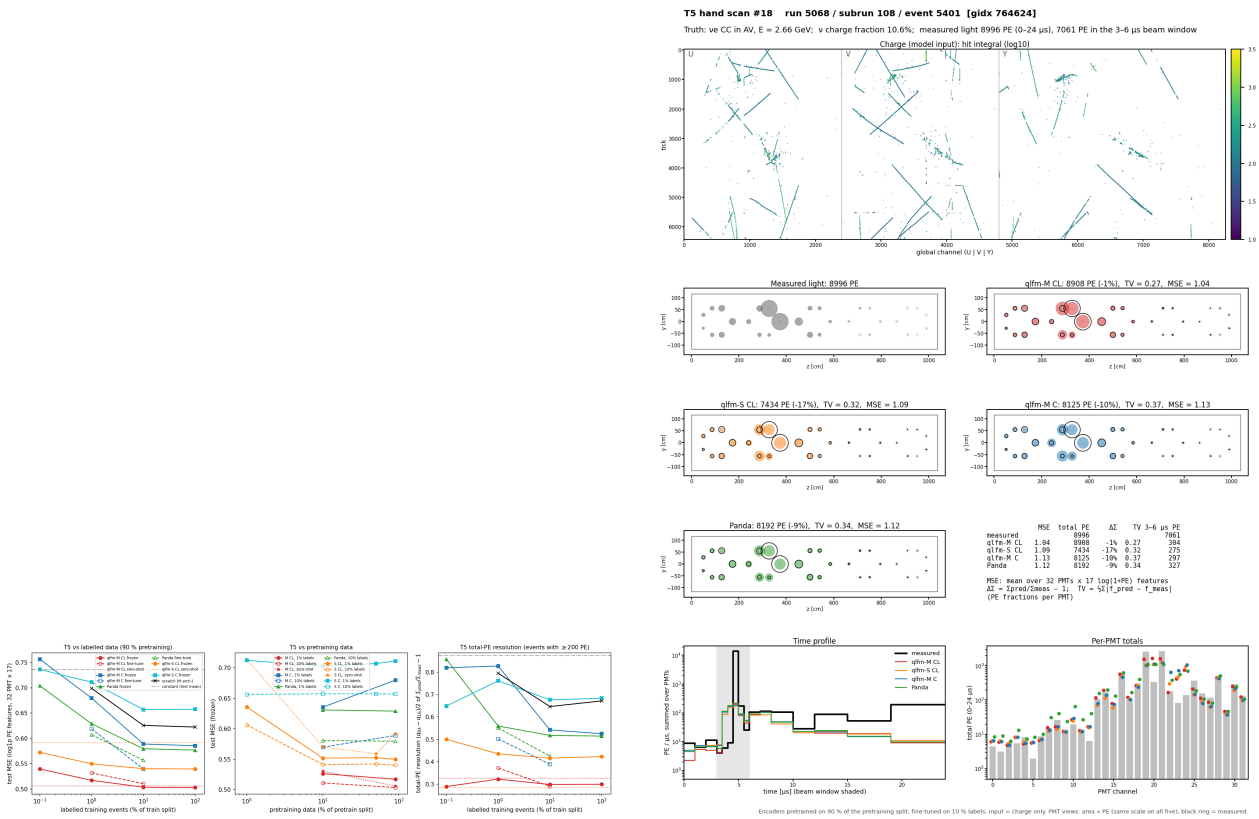


Figure 4: PLACEHOLDER (TN4). Left: T5 test MSE vs pretraining and label fraction. Right: one hand-scan event (measured vs predicted PMT light). Final: T5 bar chart for the four models + zero-shot, and one display.

3.4 Multi-task fine-tuning

[**TODO:** New. One backbone, five heads; loss per task masked to its eligible events, weights normalized by each task's initial loss. Compare with single-task fine-tunes at 0.1% and 1%: how much does sharing the backbone cost?]

Table 3: PLACEHOLDER layout. Single-task (ST) vs multi-task (MT) fine-tune, qlfm-M CL. Values pending.

Labels	Fine-tune	T1	T2	T3	T4	T5
0.1%	ST / MT	– / –	– / –	– / –	– / –	– / –
1%	ST / MT	– / –	– / –	– / –	– / –	– / –

4 Which multi-modal FM architecture suits LArTPCs?

So far only qlfm saw light in pretraining; the references got light only at the head. Other recipes may absorb the charge–light correlation better. We give each recipe light in pretraining and compare at matched size (≈20M) and compute (20 node-hours of pretraining).

Table 4: The four recipes. Architecture and objective change together except for the MAE/JEPA pair, which isolates the objective on one backbone.

Recipe	Backbone	Objective	How light enters pretraining
qlfm-M CL	Transformer	MAE (charge + light)	PMT tokens in the sequence (as now)
qlfm-M CL JEPA	Transformer	JEPA (I-JEPA-style)	same tokens; predictor regresses EMA-target embeddings of masked charge blocks and masked PMTs
PTv3 + Sonata + L	PTv3	self-distillation	event-level cloud: 3 planes plus 32 PMT points in a fourth slab, placed at their (z, y); per-modality input embedding [**TODO:** verify in smoke test]
Mamba-2 + L	Mamba-2 (FM4NPP m4)	k-NNN + masked light	PMT tokens as prefix (conditioning) and suffix (masked light regressed from the charge); hits as in FM4NPP

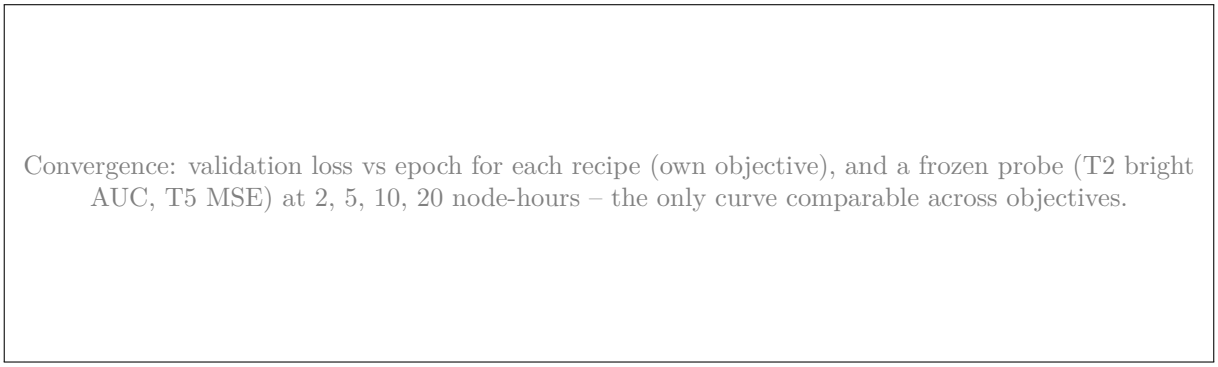


Figure 5: PLACEHOLDER Pending (Appendix C).

Table 5: PLACEHOLDER layout. Multi-task fine-tune at 1% labels (selection metric), and frozen at 1%; 3 seeds.

Recipe	T1	T2	T3	T4	T5	mean rank
qlfm-M CL (MAE)	–	–	–	–	–	–
qlfm-M CL (JEPA)	–	–	–	–	–	–
PTv3 + Sonata + L	–	–	–	–	–	–
Mamba-2 + L	–	–	–	–	–	–

5 How the correlations are learned

- **Embedding maps.** t-SNE / PCA of frozen event and token embeddings without a learned map, coloured by ν/cosmic, flavour, energy, drift coordinate (as FM4NPP Fig. 8 [6]).
- **Probes.** Linear probes on charge tokens only (light masked at inference) for the drift coordinate x (which needs t_0), deposited energy and the vertex: CL vs C encoder.
- **Fusion depth** (survey E1). Light enters at the input, after $L/2$ blocks, or only at the head, all at matched compute.
- **Oracle 3D input** (appendix-level). The open data has no space points; a 3D input can only be truth energy deposits. It gives an upper bound on what 3D reconstruction could add, and is labelled as an oracle.

- **Light-swap counterfactual** (survey E2, optional): replace an event's light by another event's and measure the drop.

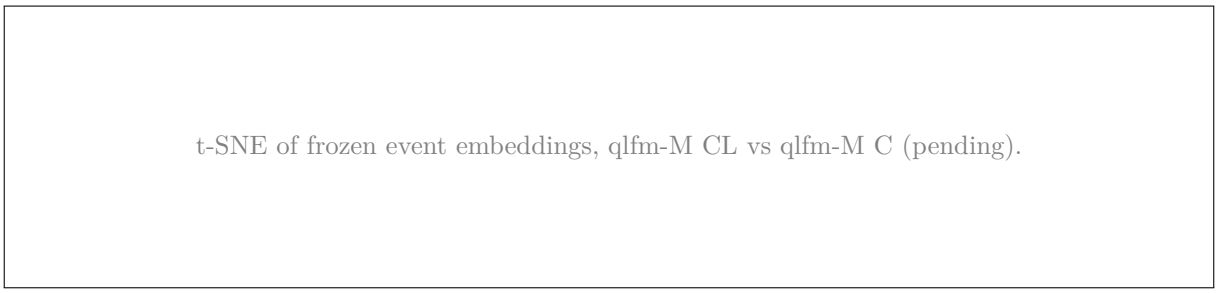


Figure 6: PLACEHOLDER Pending.

6 Scaling

On hold until Sections 3–5 are done and the go-ahead is given. Scaling of each recipe with model size and compute; smaller charge cells (more tokens), where SSMs may gain (Josh's SSM survey).

7 Discussion and limitations

Overlay simulation (no beam-on data); one detector; models of about 20M parameters; compute-limited training; T4 on a clean subset. [**TODO:** expand.]

A Model setups

[**TODO:** One subsection per recipe: tokenizer, backbone, objective, hyper-parameters, pretraining compute, heads; MAE input/output and token figures from TN1 (Figs. 2–4).]

B Event displays

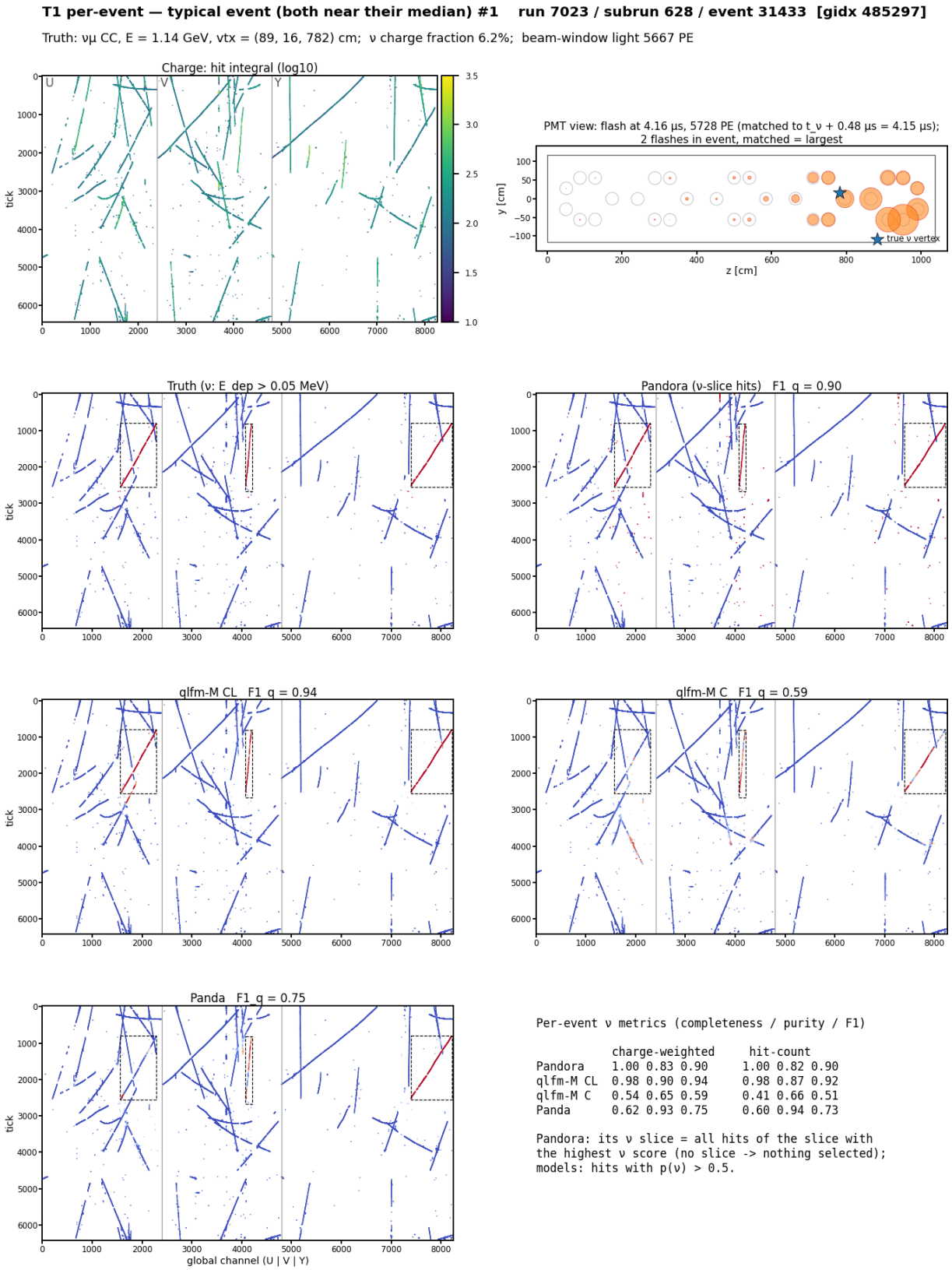


Figure 7: PLACEHOLDER (TN4). A typical event: T1 hit predictions of qlfm-M CL, qlfm-M C, PTv3 and Pandora.

C Campaign plan

C.1 Decisions so far (6 Nov 2026)

- Section 4 first; total budget ≤200 node-hours (nh), charged to `neutrinoGPU::WireCellDev`, preemptable queue.
- All four recipes pretrained at the same budget, 20 nh each, including a new qlfm-M CL run (the existing one used about 2 nh, the round-2 references about 4 nh).
- T4 scored on a truth-based subset common to all models (not on each model's T1).
- JEPA on the qlfm Transformer, so MAE vs JEPA isolates the objective.
- "Reco-3D" becomes an oracle study with truth deposits (no space points in the open data).
- Naming: PTv3 + Sonata ("as used in Panda"), not "Panda" and not "PT2".
- Venue: ICML 2027 main conference, *Application-Driven ML* area (deadline expected late January 2027); fallbacks: an ICML 2027 workshop, the NeurIPS 2027 Datasets & Benchmarks track, MLST.

C.2 Section 4 job plan

Step 0 – code (login node + short debug jobs).

1. Multi-task fine-tune mode in `train_downstream` (five heads, masked per-task losses, loss normalization); works for all backbones through the common interface.
2. Light for PTv3: event-level clouds (3 planes + PMT points), Sonata views include PMTs; pretraining config `ptv3_sonata_cl`.
3. Light for Mamba-2: PMT prefix and suffix tokens, masked-light regression added to the k-NNN loss.
4. JEPA objective on the qlfm backbone (context encoder, EMA target encoder, narrow predictor; block masks on each plane plus masked PMTs); collapse monitor (embedding rank).
5. Logging: validation loss every ≈15 minutes; checkpoints kept at 2, 5, 10, 20 nh for the probe curve.

qlfm-M CL needs no new code and starts first.

Step 1 – smoke tests. 30-minute runs of each new recipe: loss falls, no NaN, throughput measured (sets the step count for 20 nh), resume after a kill works.

Step 2 – pretraining. Four 1-node preemptable jobs of 20 h each (`-r y`, resumable), running in parallel; the feeder keeps at most 6 jobs in the queue.

Step 3 – downstream. For each recipe, using the 1-node `queue.pbs` worker (4 tasks per node):

- frozen heads, T1–T5, at 0.1% and 1%;
- multi-task fine-tune at 0.1% and 1%, 3 seeds (selection metric: 1%);
- single-task fine-tunes of qlfm-M CL at 1% (for Table 3);
- probe curve: frozen T2 and T5 at 1% on the 2/5/10/20 nh checkpoints.

Table 6: Budget for Section 4 (node-hours). Downstream costs from rounds 1–4: a frozen or single-task head run takes 0.1–0.3 nh per GPU slot (4 per node); PTv3 fine-tunes are about 2× slower.

Item	per recipe	total	note
Code and smoke tests	3–4	13	debug queue
Pretraining	20	80	preemptable, 1 node × 20 h
Probe curve (4 ckpts, T2+T5)	1.5	6	
Frozen heads (5 tasks, 2 fractions)	1.5	6	
Multi-task FT (2 fractions, 3 seeds)	3–5	16	
Single-task FT (qlfm-M CL, 1%)	–	2	for Table 3
Reruns, preemption (about 25%)	–	30	
Total		**≈155**	cap 200 (45 spare)

The 45 nh spare can pay for charge-only twins of the new recipes or for the fusion-depth ablation of Section 5.

C.3 Timeline (target)

Week of 9 Nov	Step 0–1; qlfm-M CL 20 nh pretraining starts
16–29 Nov	PTv3 + L, Mamba-2 + L, JEPA pretraining
30 Nov – 13 Dec	downstream, probe curves; Section 4 results
14 Dec – 10 Jan	Section 3 updates (metrics, multi-task FT), Section 5
11–27 Jan	writing; ICML deadline (expected late January)

C.4 Risks

- **PTv3 + L design** (PMT points in a point-cloud backbone) and Sonata's view augmentations may not suit PMT points; fall back to a light cross-attention adapter inside the encoder stages (mid fusion) and say so.
- **JEPA collapse**: our earlier self-distillation on qlfm tokens (qlfm-SD) underperformed, probably due to a low learning rate; monitor embedding rank, tune the LR in the smoke test.
- **Mamba-2 and masking**: SSMs fit MAE poorly (Josh's SSM survey); we keep FM4NPP's k-NNN objective and add masked light at the suffix.
- **Preemption**: runs resume from checkpoints; validated only with a normal job so far.

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